

# LD9 Final Project Summary

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*Final Documentation Package for the Bank Marketing Subscription Prediction Project*

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# Learning Day 9 — Ensemble Methods Project Flow

What we completed from Notebook 1 to Notebook 6

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## Notebook 1 — Source Audit & Plan

- Audited the uploaded course files
- Built a coverage matrix for all required topics
- Selected the Bank Marketing dataset
- Defined the Day 9 notebook roadmap and evaluation plan

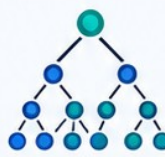
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## Notebook 2 — Data Preparation & Baselines

- Explored target balance, missing values, and feature types
- Created train/test split and preprocessing pipeline
- Trained baseline models: Dummy, Logistic Regression, Decision Tree, KNN
- Saved reusable data and baseline outputs

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## Notebook 3 — Bagging & Random Forest

- Compared Single Decision Tree vs Random Forest (100 trees)
- Reviewed feature importance for both models
- Tested  $n_{\text{estimators}}$  from 10 to 500 and found stagnation around 100
- Checked OOB score and Random Subspace Method via  $\text{max\_features}$
- Confirmed Random Forest clearly outperformed a single tree

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## Notebook 4 — Cross-Validation & StackingClassifier

- Ran 5-fold cross-validation for the base learners
- Built a StackingClassifier using RF, Logistic Regression, KNN, and Decision Tree
- Compared Logistic Regression vs Random Forest meta-models
- Studied model diversity and compared CV + held-out test results
- Best held-out stacking model: RF meta-model

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## Notebook 5 — Manual OOF Stacking & Blending

- Built manual 5-fold Out-of-Fold prediction features
- Trained a meta-model on OOF meta-features
- Built Blending with a 70/30 split
- Compared OOF Stacking vs Blending in performance and stability
- Found Manual OOF Stacking stronger than Blending

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## Notebook 6 — Three-Layer Stacking & Gradient Boosting

- Created a 3-layer stacking system: Level 0, Level 1, Level 2
- Evaluated the final Three-Layer Stacking model
- Added a compact Gradient Boosting model for theory-to-practice coverage
- Compared complexity vs business value
- Practical recommendation: simpler RF-meta stacking remained the best choice



## Outcome So Far

- ✓ All major Day 9 ensemble topics from Notebook 1 to 6 were covered
- ✓ Practical methods included Bagging, Random Forest, OOF Stacking, Blending, Three-Layer Stacking, and Gradient Boosting
- ✓ Best practical model so far: StackingClassifier with Random Forest meta-model

*Figure: High-level workflow from Notebook 1 to Notebook 6.*

## 1. Project Objective

Build and compare ensemble learning models for the bank marketing dataset, covering Bagging, Random Forest, Stacking, Blending, Three-Layer Stacking, and Gradient Boosting, while fully satisfying the Learning Day 9 course scope and MVD requirements.

## 2. Dataset and Prediction Target

- Dataset: Bank Marketing dataset chosen for a business-relevant binary classification task.
- Target: Predict whether a customer subscribes to a term deposit.
- Project fit: Strong match to CRM, campaign optimization, BI reporting, and classification use cases.

## 3. Project Workflow

| Notebook | Main focus                        | Main deliverable                                     |
|----------|-----------------------------------|------------------------------------------------------|
| 1        | Audit and planning                | Coverage matrix and selected dataset                 |
| 2        | Data preparation and baselines    | Preprocessing pipeline and baseline benchmark        |
| 3        | Bagging and Random Forest         | Tree vs forest comparison and RSM/OOB analysis       |
| 4        | StackingClassifier                | CV comparison and meta-model comparison              |
| 5        | Manual OOF stacking and blending  | Custom stacking logic and stability comparison       |
| 6        | Three-layer stacking and boosting | Complexity-vs-value decision and boosting connection |
| 7        | Final proof notebook              | MVD checklist and final recommendation               |

## 4. Best Final Results

| Model                        | Accuracy | Precision | Recall | F1     | ROC-AUC |
|------------------------------|----------|-----------|--------|--------|---------|
| StackingClassifier (RF meta) | 0.8598   | 0.8176    | 0.9064 | 0.8597 | 0.9247  |
| Three-Layer Stacking         | 0.8612   | 0.8369    | 0.8781 | 0.8570 | 0.9251  |
| Manual OOF Stacking          | 0.8603   | 0.8354    | 0.8781 | 0.8562 | 0.9259  |
| StackingClassifier (LR meta) | 0.8589   | 0.8338    | 0.8771 | 0.8549 | 0.9265  |
| Blending 70/30               | 0.8554   | 0.8314    | 0.8715 | 0.8509 | 0.9232  |
| Gradient Boosting            | 0.8325   | 0.8109    | 0.8431 | 0.8267 | 0.9123  |

The recommended final model is the StackingClassifier with Random Forest meta-model because it gave the best overall balance for the business objective.

## 5. Important Findings

- Random Forest clearly outperformed a single unpruned Decision Tree.
- Feature importance results were broadly aligned between the tree and the forest, with duration, balance, age, day, and outcome/contact-related variables playing important roles.
- Random Forest performance improved quickly and then stagnated around 100 trees.
- Stacking improved over most individual base learners.
- Manual OOF stacking worked almost as well as automatic stacking and was valuable for learning the inner logic of stacking.
- Blending was simpler but slightly weaker and more split-dependent.
- Three-layer stacking was technically successful but not sufficiently better to justify the complexity.
- Gradient Boosting was useful conceptually but weaker than the best stacking models on this dataset.

## 6. Final Recommendation

Choose the RF-meta StackingClassifier for the final solution. It gives the strongest F1-score and the best recall, making it highly suitable for subscription targeting.

## 7. Limitations and Next Steps

- Threshold tuning was not the main focus; this could further improve business alignment.
- Explainability tools such as SHAP could be added later for richer model explanation.
- A production version would include monitoring, retraining schedule, and scoring pipeline automation.